

Multi-Camera Hand Motion Analysis For Puncture Technique Training

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Abstract—Injection skill training is a crucial element in the clinical medical curriculum for medical students. However, existing feedback methodologies are predominantly based on subjective evaluations by physicians, which lack critical objective measures for assessing the precision and dependability of injection technique. Additionally, implementing wearable sensors is exorbitantly costly, and their usage can negatively influence the students' movements and performance due to the discomfort and encumbrance they cause. To address these challenges, our research presents a novel MediaPipe Hand-based system for non-contact tracking of finger movements during simulated injection training. This system employs four sets of cameras positioned at various angles to track medical students' injection movements, effectively resolving the complex issue of finger occlusion. It also reconstructs a 3D model of the hand during the injection process. To ascertain the accuracy of our system, we used a high-precision motion capture system (VICON) as a standard. Data on hand injection techniques were gathered from ten medical students. The system's robustness was verified by comparing the injection finger joint lengths, while the system's precision was evaluated by contrasting the finger joints' positions. The methodology proposed in this study can provide objective, accurate feedback on the trainees' techniques to enhance their injection skills and elevate the standard of clinical injection training.

Index Terms—Clinical injection training, MediaPipe hand, Non-contact, 3D reconstruction

I. INTRODUCTION

Injection proficiency is an essential component of the clinical training regimen for medical students. Mastery of this technique is crucial for ensuring patient safety and effective treatment while minimizing unnecessary harm to patients [1]. Traditionally, the training and evaluation of injection skills have largely depended on the subjective assessments of experienced physicians [2]. Although this approach benefits from the meticulous understanding of seasoned practitioners, it inherently lacks the quantitative analysis crucial for accurately measuring the precision and reliability of students' injection techniques [3], and also imposes additional workload on physicians.

To tackle these challenges, researchers employ wearable sensors for motion tracking and assessment. A prototype system developed by [4] captures finger position data during syringe grasping, facilitating injection practice with a finger motion capture system. Recent studies emphasize the role of wearable sensors in medical training and rehabilitation [5]. However, wearables have inherent limitations. In scenarios requiring intricate hand movements, sensor placement can cause discomfort and hinder natural motion, leading to suboptimal training experiences [6]. Additionally, the cost of these sensor-based solutions can be relatively high.

Virtual Reality (VR) technology has been utilized to simulate injection skill training, offering innovative educational solutions [7]. Studies [8] demonstrate that VR simulation significantly enhances nursing students' knowledge, confidence, and clinical proficiency in IV injection training. Furthermore, [9] underscores the educational benefits of integrating traditional arm models with VR and haptic technology for first-year nursing students' intravenous simulation training. This approach effectively addresses the limitations and challenges associated with wearable sensors. While these technologies provide a safe learning environment, reducing stress and boosting confidence, they may not fully replicate the complexity of real-world injection techniques and patient interactions.

Recognizing these challenges, our study introduces an innovative non-contact hand motion tracking system for simulated injection actions, leveraging Mediapipe hand tracking technology [10]. This system utilizes industrial cameras to capture medical students' hand movements during injection exercises. Unlike single-camera setups, this approach mitigates finger occlusion issues and allows students to practice naturally without physical constraints. Consequently, it offers a more realistic training experience that mirrors clinical procedures.

The subsequent section of the paper elaborates on our methodology, encompassing the hardware construction, hand model reconstruction, occlusion solutions, and details of the

evaluation experiments. The paper culminates with an in-depth assessment of the method's accuracy and reliability, followed by a discussion and conclusion.

II. METHODOLOGY

This section delineates the hardware system's construction, the hand model's reconstruction, and the solutions implemented for occlusion issues.

A. Construction of Hardware System

We devised an original multi-camera tracking system to analyze hand movements during injection. Based on the results of multiple experiments and considering the diversity of the subjects and the variations in injection movements, we have identified the optimal placement of cameras for the experiments, as shown in Figure 1:

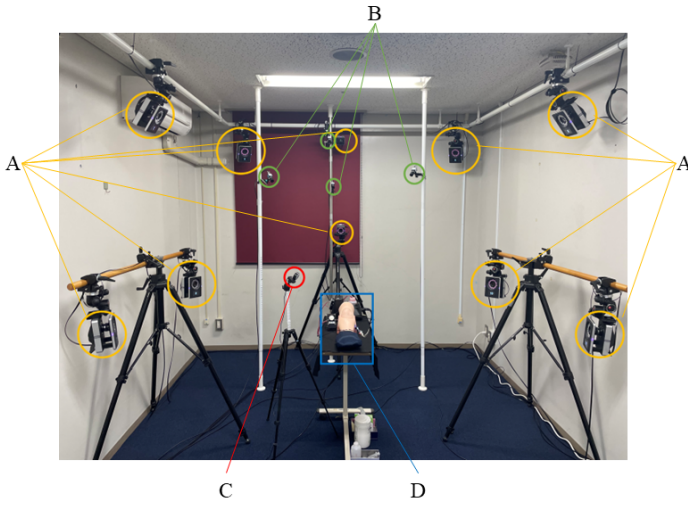


Fig. 1. The construction of Multi-camera tracking system.

Figure 1 presents the positioning of the designed multi-camera tracking system during the experiments. (A) represents the camera locations used by the high-precision reference system VICON in this study. During the experiments, it synchronously tracks hand movements during injections and is subsequently compared to the results obtained by the proposed method in this paper. (B) denotes the positions of four industrial cameras employed in this research for tracking hand movements during the injection process. (C) signifies the camera locations for measuring temporal parameters during hand injection. The correctness of the injection is determined by detecting the length of the needle inserted into the skin [12] using YOLOv7 technology [13]. (D) describes the simulated hand injection model used by medical students during the operation.

B. Reconstruction of the hand model

To reconstruct the hand model, we employed Mediapipe hand algorithm for real-time tracking of both hands during injection [10]. Mediapipe Hand is capable of distinguishing between left and right hands and can identify a total of 42

landmarks based on finger joints, providing 2D coordinates for each. This is illustrated in Figure 2.

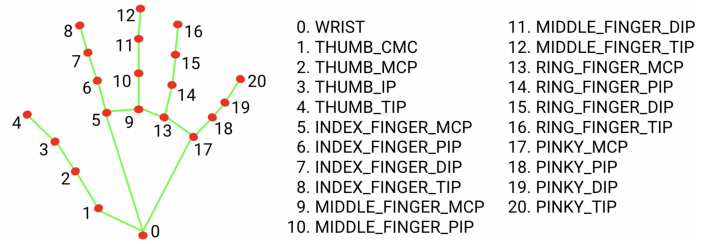


Fig. 2. The Mediapipe hand landmarks position [10].

First, we needed to calibrate the four cameras. This was done by capturing multiple sets of images with a calibration board, which helped in obtaining the internal parameters and distortion coefficients of the cameras [11]. Next, images of the calibration board were simultaneously taken with all four cameras to complete the stereo calibration. This process involved determining the origin of the world coordinate system through any point on the chessboard grid.

After completing the calibration process, we were able to obtain 2D landmark coordinates based on Mediapipe hand tracking from the perspective of all four cameras. In this study, we employed the Direct Linear Transformation (DLT) and Singular Value Decomposition (SVD) algorithms to transform these points from the camera coordinate systems into the world coordinate system, enabling 3D hand model reconstruction.

- **Direct Linear Transform (DLT).** The DLT algorithm is efficient for computing world coordinates from image points. It is particularly useful when dealing with multiple views, as it recovers the projection mapping relationship between these views. In the context of the DLT method, let's assume we have sets of points U_1 and U_2 derived from two different perspectives, corresponding to image points in cameras A and B, respectively. Each point in U_1 and U_2 can be associated with a world coordinate point X through the projection matrices P_1 and P_2 of cameras A and B. The DLT method utilizes these points to establish a system of linear equations. By solving this system, the 3D coordinate point X can be determined.

$$\begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} P_1 \\ P_2 \end{bmatrix} X \quad (1)$$

- **Singular Value Decomposition (SVD).** Singular Value Decomposition (SVD) is an effective tool for handling overdetermined systems, where there are more equations than unknowns. In our case, with four cameras in the setup, we have more equations than unknowns. We construct a matrix A , derived from the cross-products of the projection matrices of the four cameras and the image points. SVD is used to decompose A into U , Σ , and V^T , where Σ is a diagonal matrix composed of singular values, and both U and V are orthogonal matrices.

$$A = U\Sigma V^T \quad (2)$$

The least squares solution to the linear system $AX = 0$ can be obtained by computing the pseudo-inverse of matrix A . With SVD, we can find the pseudo-inverse A^+ of matrix A , and then calculate the least squares solution:

$$X = A^+B \quad (3)$$

Where B is a vector comprised of the coordinates of the corresponding image points.

For the 3D reconstruction of the hand model in my research, we start by gathering 2D landmark coordinates from four camera perspectives. Then, we utilize the DLT algorithm to calculate the corresponding 3D world coordinates. Finally, by applying SVD to handle the overdetermined system arising from data collected from multiple cameras, we obtain a precise set of 3D coordinate points. These points represent the positions of key hand landmarks in 3D space.

C. Occlusion problem

After obtaining the reconstructed 3D hand model, it is inevitable to address the issue of hand occlusion caused by complex hand movements. Naturally, occlusion problems can be categorized into two main types:

- **Confidence distance (Error problem).** Due to occlusion issues that result in unidentifiable landmarks, we resolve this through confidence distance calculations. Firstly, our proposed system employs four cameras, and any combination of at least two of them yields 11 sets of results. Consequently, for any given landmark, we can derive 11 sets of predicted 3D coordinates.
 - a. Calculate the distances between different camera prediction points using the Euclidean distance formula:

$$d_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2 + (z_j - z_i)^2} \quad (4)$$

where x_i, y_i, z_i represent the coordinates of point i , and x_j, y_j, z_j represent the coordinates of point j .

- b. Compute similarity scores using the Gaussian similarity function:

$$S_{ij} = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \quad (5)$$

- c. Aggregate similarity scores for each camera to determine the weights. d. Normalize the weights. e. Employ the weights as a threshold to eliminate errors and output the median value as the correct result.

- **Smooth Handling (Shake problem).** To address the issue of Mediapipe hand jitter caused by occlusion, we propose the use of smoothing techniques as a solution. After conducting multiple comparative experiments, we have determined that Gaussian smoothing is most suitable for our research, with the optimal smoothing parameter being set at 21.

III. EXPERIMENT AND RESULTS

Our dataset was sourced from grades 4-6 students at Chiba University. Ultimately, we curated ten injection datasets for this study.

To validate our proposed system, we compared our calculated results with those from the VICON reference system. We employed landmarks at the same positions as those used by Mediapipe hand, as depicted in Figure 3:

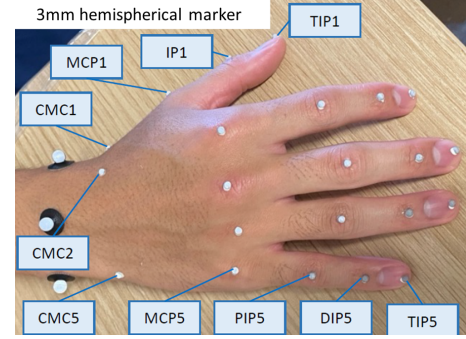


Fig. 3. The VICON system marker position.

Tracking accuracy, stability, and robustness are the primary indicators for assessing system performance.

A. Evaluation of stability/robustness (Joint Lengths)

To assess the stability and robustness of the system, we select the intrinsic attributes of the fingers (joint lengths) whenever possible to validate the proposed method. As illustrated in Figure 4, the labels of the finger joints are indicated:

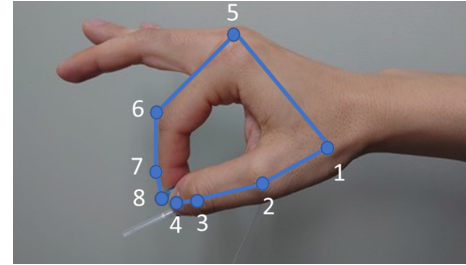


Fig. 4. Label of the finger lengths.

For measurements of the same joint length, as shown in Figure 5, the results are shown for the joint lengths measured frame by frame during the injection training. We also compared the results across multiple sets of experimental subjects, and the findings revealed the following ranking: Multi-camera (Four cameras) > Occlusion 1 (Three cameras) > Occlusion 2 (Two cameras).

TABLE I
ERROR MEASUREMENTS FOR DIFFERENT CAMERA CONFIGURATIONS

Configuration	Mean Error (mm)	Standard Deviation (mm)
Occlusion 2	5.40	4.65
Occlusion 1	4.77	4.30
Multi-camera	3.88	3.86

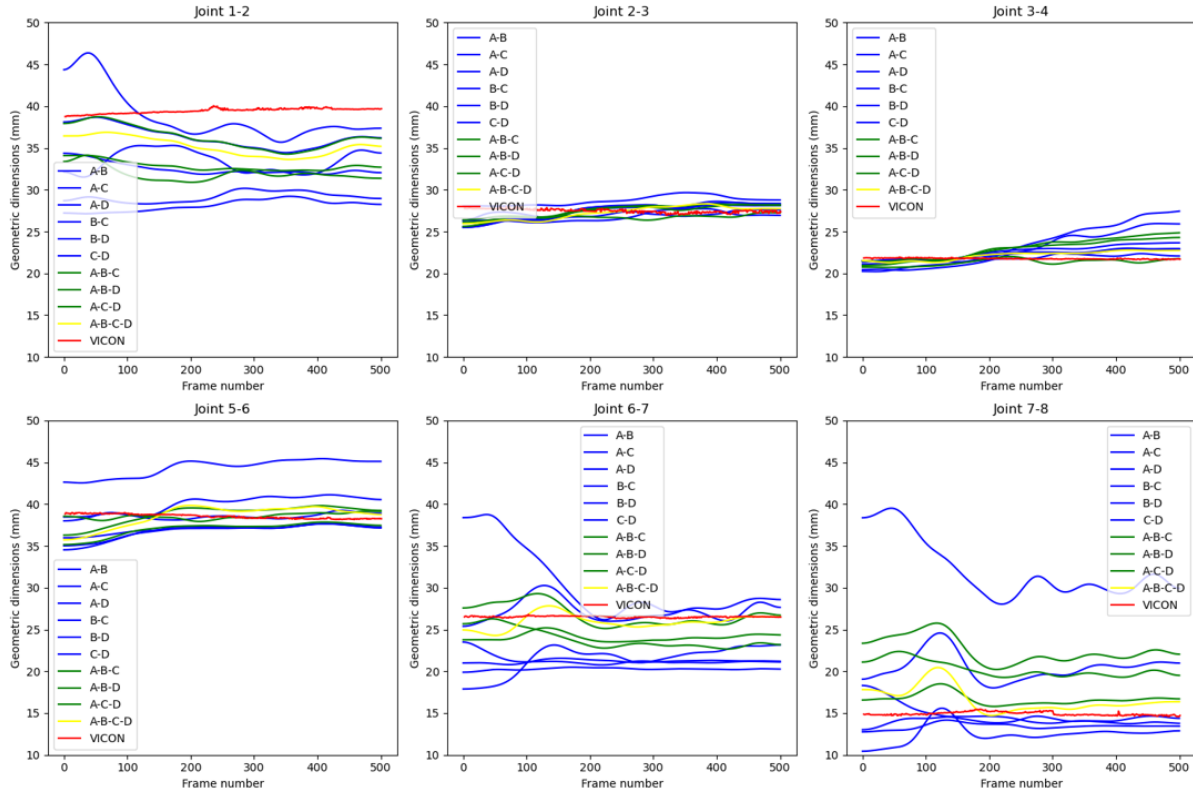


Fig. 5. Changes in the length of each joint during injection.

From the table I, it can be concluded that the "Multi-camera" configuration outperforms "Occlusion 1" and "Occlusion 2" in measuring hand joint lengths, exhibiting lower average errors and standard deviations. This indicates that our proposed method is more reliable and accurate when using four cameras.

B. Evaluation of tracking accuracy (Joint Position)

To further validate the accuracy of our system, we aligned the coordinate systems of both systems to compare the differences in joint positions between our proposed method and the VICON system. As shown in Figure 6.

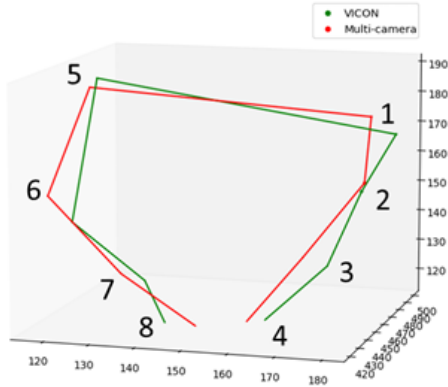


Fig. 6. The joint positions in different world coordinate systems are aligned using the least squares method.

We conducted a quantitative analysis by calculating the root mean square error (RMSE) of the positions of eight joints in each frame. The visualized results are shown in the figure 7.

In this visualization (Figure 7), the point position represents the trajectory of the joint's movement based on the ground truth. We used color mapping to represent distance errors, with blue indicating smaller errors and red indicating larger errors. Subsequently, we compared the RMSE errors of joint positions for 10 objects, and the results are presented in the table II:

TABLE II
RMSE ERRORS FOR JOINT POSITIONS

Joint Point	Mean $\pm$ Std (mm)
Point 1	10.78 $\pm$ 4.71
Point 2	10.79 $\pm$ 5.61
Point 3	9.22 $\pm$ 5.51
Point 4	9.92 $\pm$ 5.49
Point 5	7.19 $\pm$ 2.92
Point 6	7.59 $\pm$ 4.47
Point 7	6.70 $\pm$ 2.49
Point 8	11.70 $\pm$ 6.57

Joints 5, 6, and 7 exhibit lower mean RMSE errors and relatively low variability, indicating more accurate and consistent measurements. On the other hand, Joints 1 to 4 demonstrate moderate to high errors and variability, while Joint 8 displays the highest level of error and variability.

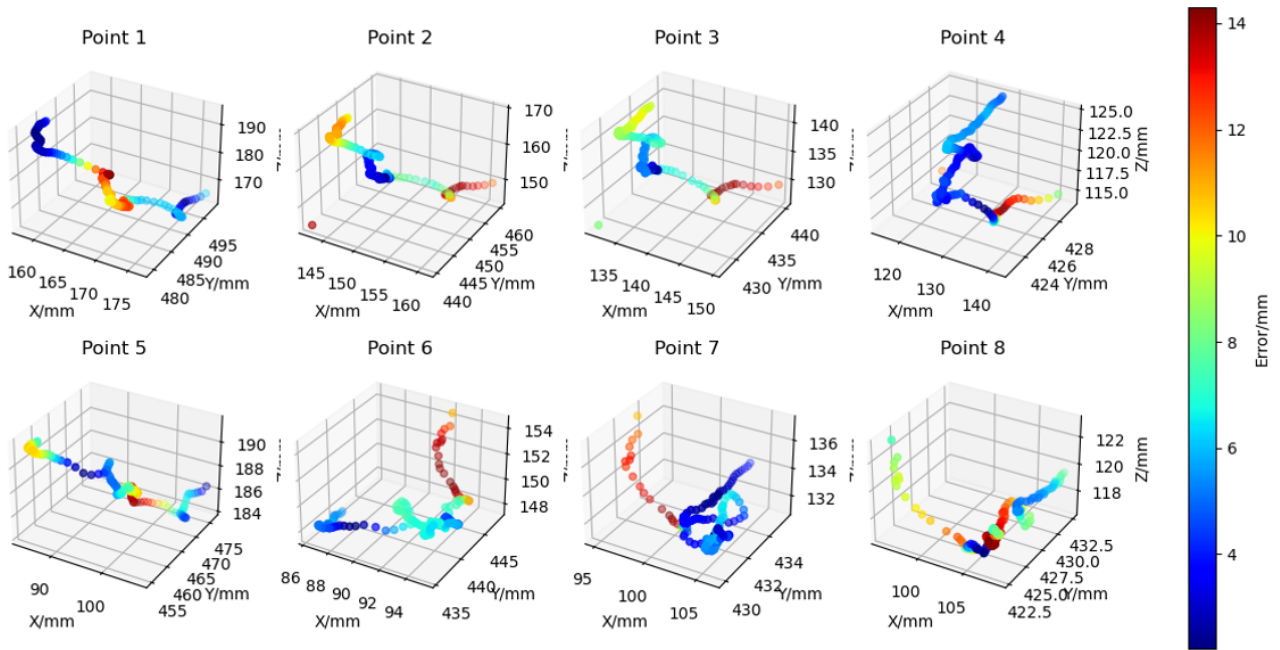


Fig. 7. Comparison of the RMSE of joint positions for the (thumb-index finger) joint in the same world coordinate system based on the injection finger.

IV. DISCUSSION AND CONCLUSIONS

The study presents an innovative method that overcomes limitations associated with wearable sensors and occlusion challenges, addressing the complexities of assessing injection skills. Here is a summary of this paper:

Traditional subjective assessment methods rely on experienced medical practitioners, lacking the objective metrics for precision. Thus, the paper introduces a non-contact hand-based simulated injection tracking system, utilizing multiple cameras and Mediapipe hand technology. This system allows medical students to practice injection techniques without sensors, providing a more natural training experience. Additionally, the paper tackles issues related to hand occlusion during tracking with two key solutions: Confidence Distance for error correction and Smooth Handling for shake reduction, which significantly improves tracking accuracy. The study conducts experiments with a dataset of medical students, rigorously comparing the results of the proposed system to a high-precision reference system, VICON. The results affirm the precision of the proposed system.

This paper introduces an innovative approach to enhance medical students' injection skills training. By addressing the limitations of traditional methods and wearable sensors, it presents a non-contact hand-tracking system. The paper validates the system's accuracy and robustness through experiments, laying the groundwork for quantitative research in medical education.

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